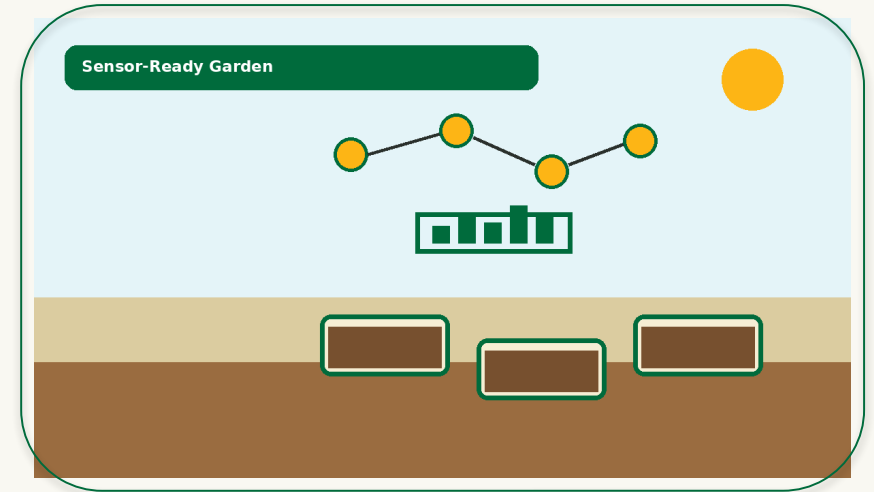


Sensor-Ready Garden

Introduction to Agricultural Sensing

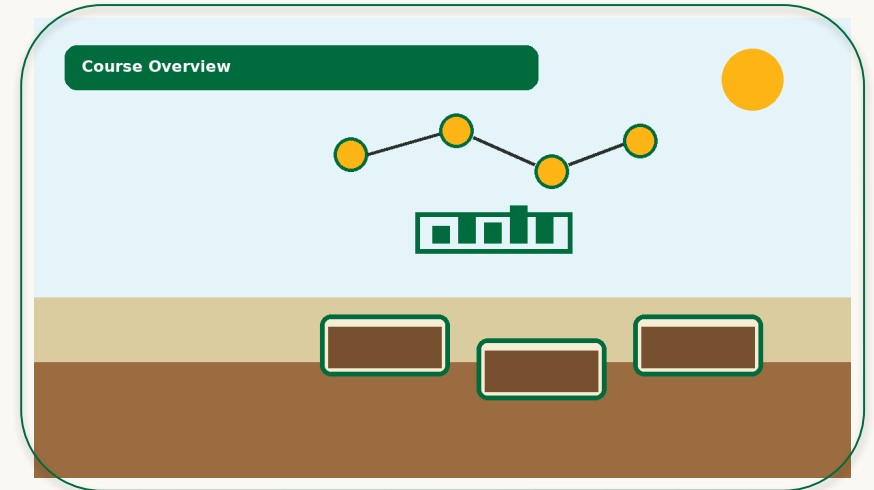
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden/laboratory system; teach all ten technical topics accurately; guide students in measurement, comparison and data collection; connect agricultural science with mathematics and systems reasoning; supervise practical work safely; maintain credible records; interpret evidence without overstating conclusions.
- Core materials: teacher guide, system/plot map, approved tools and irrigation equipment, labels, clipboards, rulers/tapes, graduated containers, meters/sensors where relevant, data sheets, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, measurement protocol, experiment or systems investigation appropriate to the course.



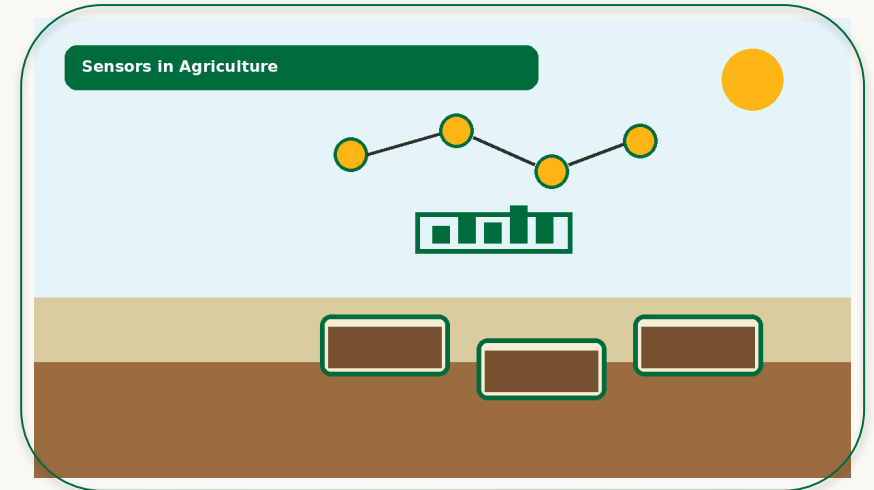
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Sensors in Agriculture

Teacher knowledge, technical practice and student supervision

- Define a sensor as a device that responds to a physical condition and produces a measurement or signal.
- Identify what each installed sensor measures and why that variable matters to crop or resource management.
- Distinguish sensor measurement from management decision.
- Explain accuracy, range and calibration conceptually.
- Use sensors as complements to field observation, not substitutes for checking the garden.



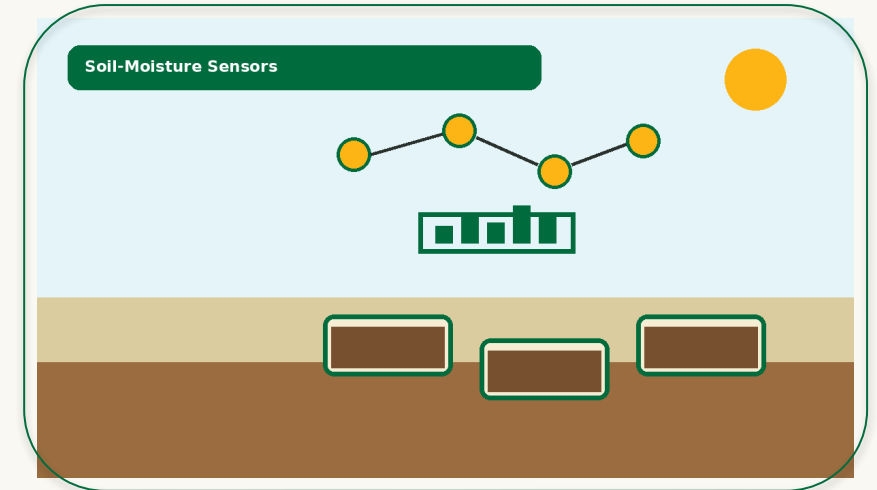
TEACHER PRACTICE

- Create a sensor-variable-decision map for the garden.

Soil-Moisture Sensors

Teacher knowledge, technical practice and student supervision

- Identify the sensor type used by the school and its measurement/output units.
- Place sensors at representative root-zone depths and positions according to manufacturer guidance.
- Avoid installing directly beside emitters unless that location is intentionally being studied.
- Compare multiple readings across a bed.
- Interpret trends rather than reacting to one isolated value.



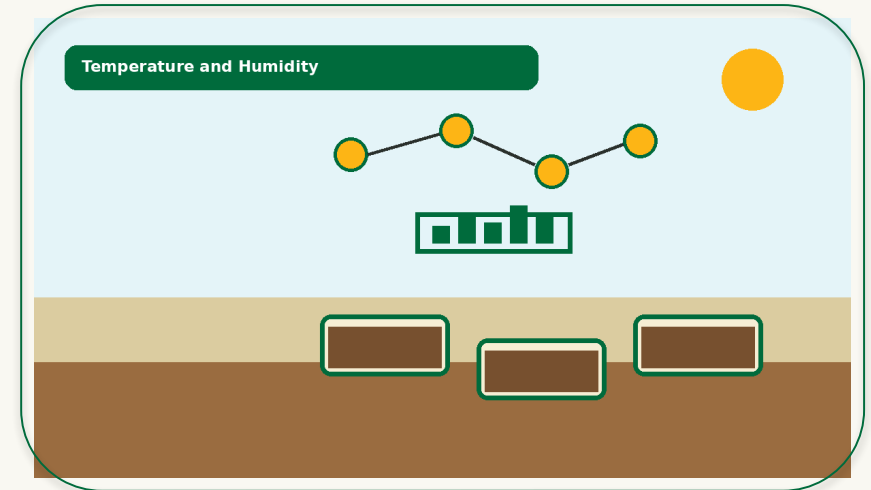
TEACHER PRACTICE

- Plan sensor positions for one irrigated bed.

Temperature and Humidity

Teacher knowledge, technical practice and student supervision

- Measure air temperature and relative humidity with shielded/appropriately placed instruments.
- Avoid direct sun, hot walls or irrigation spray unless the sensor is designed for that exposure.
- Record height and location so readings can be compared.
- Explain that temperature and humidity vary over time and space.
- Relate observations to crop conditions without claiming a single threshold applies to every crop.



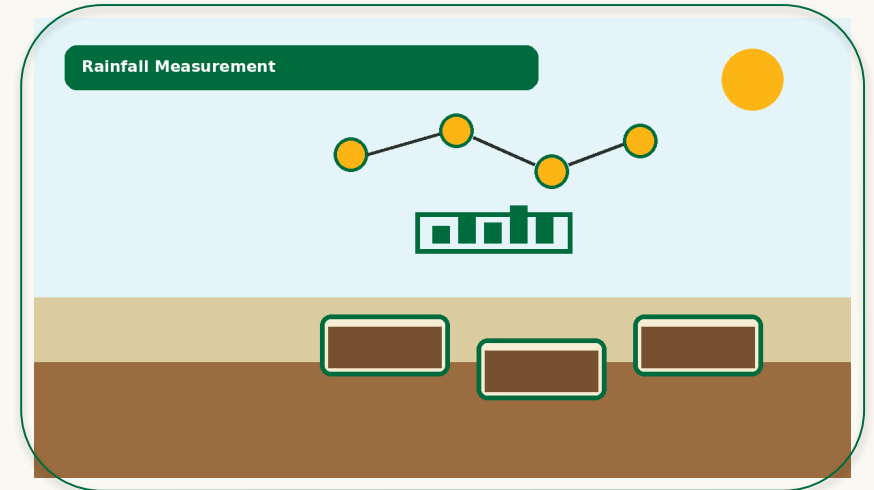
TEACHER PRACTICE

- Prepare a temperature-humidity monitoring station plan.

Rainfall Measurement

Teacher knowledge, technical practice and student supervision

- Use a properly positioned rain gauge in an open representative location.
- Keep the gauge level and away from obvious obstructions where feasible.
- Read at consistent times and record units.
- Empty manual gauges after recording according to procedure.
- Compare rainfall records with irrigation and soil-moisture data.



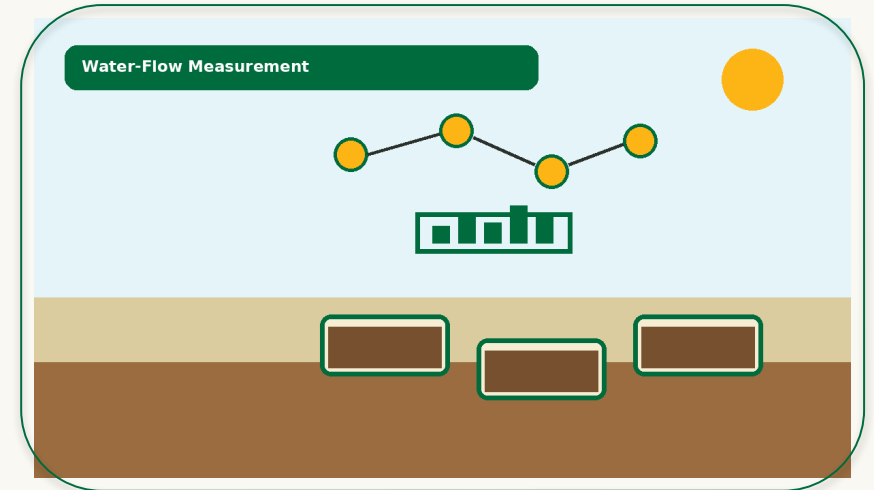
TEACHER PRACTICE

- Develop a daily rainfall record routine.

Water-Flow Measurement

Teacher knowledge, technical practice and student supervision

- Use a flow meter or timed collection to measure irrigation flow.
- Record units, irrigation zone and measurement time.
- Check for leaks or valve changes when readings shift unexpectedly.
- Distinguish instantaneous flow rate from total water volume.
- Use flow data to verify irrigation records.



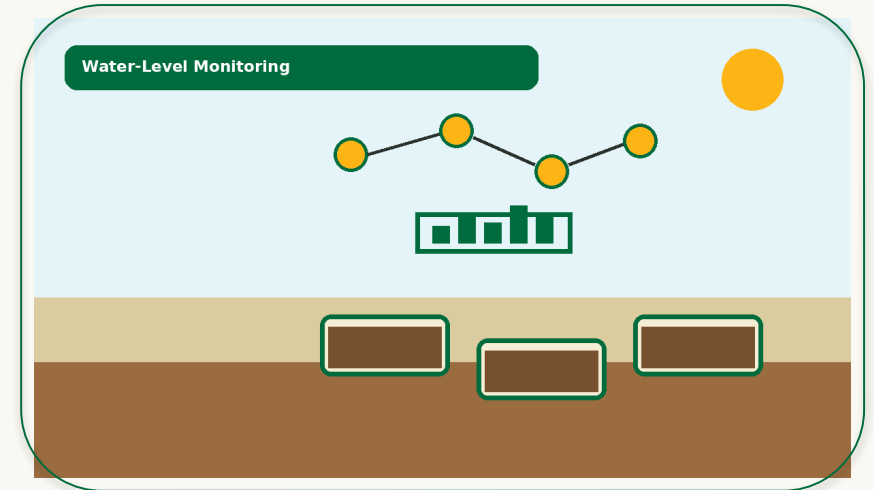
TEACHER PRACTICE

- Compare flow readings at two irrigation events.

Water-Level Monitoring

Teacher knowledge, technical practice and student supervision

- Use a level sensor, gauge or marked reservoir to track stored water.
- Define the reference point and units.
- Record levels at consistent times.
- Investigate sudden changes for irrigation use, refill, evaporation, overflow or leakage.
- Keep tanks closed and structurally secure.



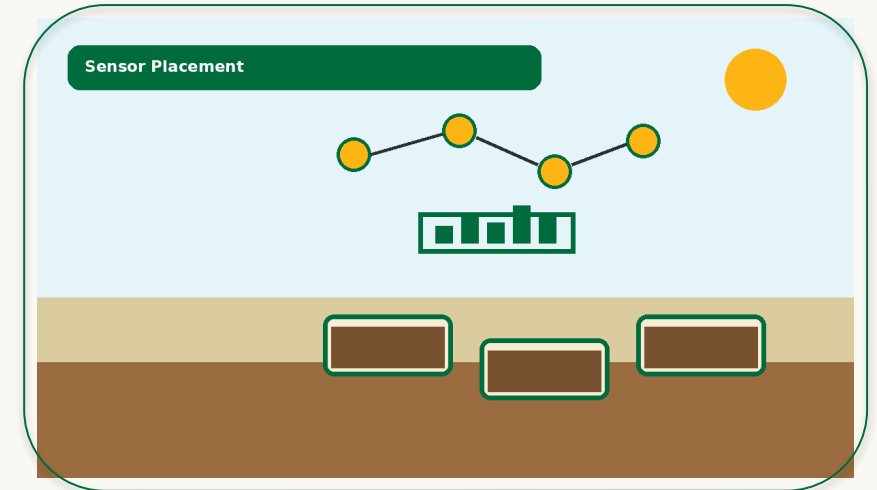
TEACHER PRACTICE

- Create a tank-level monitoring log.

Sensor Placement

Teacher knowledge, technical practice and student supervision

- Place sensors where measurements represent the question being asked.
- Avoid locations strongly biased by shade, runoff, emitters or structures unless intentionally studied.
- Document coordinates/bed position, depth and height.
- Protect cables and devices from trampling, tools and water damage.
- Reassess placement when crop canopy or system configuration changes.



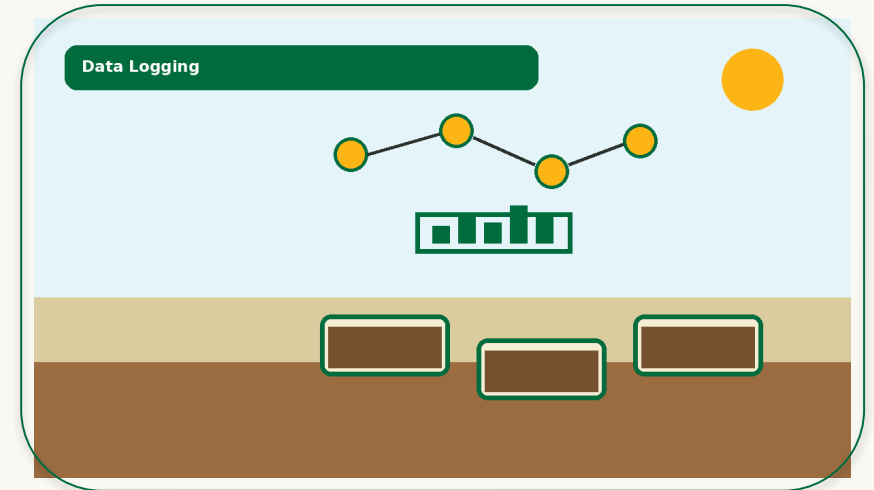
TEACHER PRACTICE

- Prepare a sensor-placement map with rationale.

Data Logging

Teacher knowledge, technical practice and student supervision

- Define data logging as systematic recording of measurements over time.
- Use consistent timestamps, units and sensor IDs.
- Check for missing values, impossible readings and clock errors.
- Back up records according to school procedures.
- Create simple time-series graphs to visualize change.



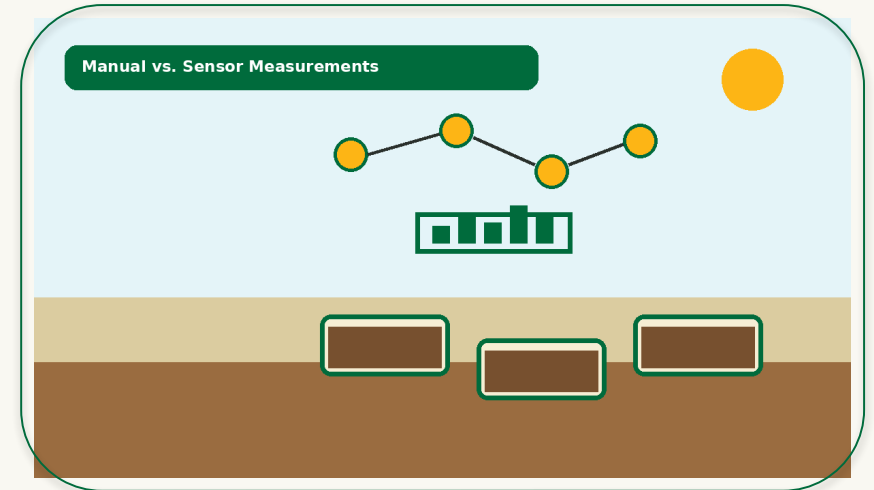
TEACHER PRACTICE

- Prepare a data dictionary and one-day logging exercise.

Manual vs. Sensor Measurements

Teacher knowledge, technical practice and student supervision

- Take paired manual and sensor measurements where a valid manual reference exists.
- Use the same location and timing as closely as practical.
- Calculate simple differences and look for systematic bias.
- Recognize that disagreement can result from method, placement, calibration or natural variability.
- Do not automatically assume the digital reading is correct.



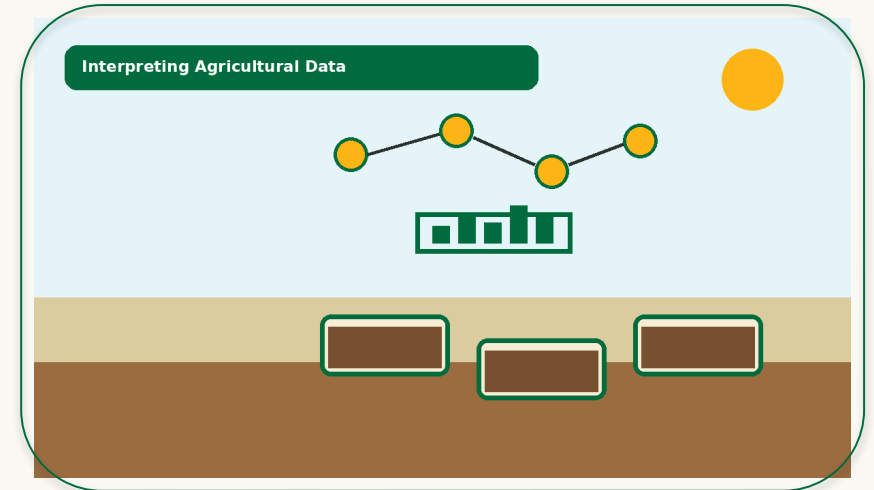
TEACHER PRACTICE

- Design a manual-versus-sensor validation check.

Interpreting Agricultural Data

Teacher knowledge, technical practice and student supervision

- Examine trends, peaks, changes and relationships among measured variables.
- Connect data to documented irrigation, rainfall and crop-management events.
- Distinguish correlation from causation.
- Flag anomalous data for verification before making decisions.
- Use evidence to formulate the next observation or management question.



TEACHER PRACTICE

- Completion task: interpret a multi-sensor garden dataset and recommend a check/action.